Autoclave is an apparatus used for sterilization of articles by steam under pressure.

- It is a metal chamber with an outer jacket and a lid, which can be firmly clamped. Steam is generated by heating water in a boiler or in the outer jacket.

In this the steam is allowed to circulate in a closed container and it is compressed and there by raises the temperature above the boiling point of water (at 121°C for 15 to 30 minute). Then the heat is turned off. The steam is evacuated.

Uses: -

The materials sterilized by autoclaving method are dressing, gloves, lines, syringes, certain instruments and culture Medias.

Points to remember:

i. All articles should be clean and dry before packing. Any organic matter such as blood or pus prevents penetration of steam.

ii. The holes in drum must be open when placing into the auto clave, and closed immediately on taking them out.

iii. Rubber gloves cannot stand high temperature and long sterilizing. Autoclave those separately at 15 lbs. pressure for 15 minutes.

iv. To auto clave bottles of fluids loosen the screw caps, evacuate the steam slowly.

Pasteurization: (Temperature at 62.8°C)

In pasteurization a high temperature is used for a short time (72°C for 15 minutes) to destroy pathogens without altering the flavor of the food. This process is used to kill all the pathogenic organisms in milk, cream, and certain alcoholic beverages.

6.6.1.3 Chemical Methods

Certain chemicals are used in disinfection of articles like thermometer and also the disinfection of floor and de-contamination of infected linen.

Chemical Substance which are commonly used:

1. Dettol: This is widely used chemical for Sterilization of instruments, thermometer etc. 5 to 50% of solution is used for drawings and wound irrigation.
2. Savlon: 1:30 solution is used to destroy or kill vegetative bacteria.
3. Chloride of lime (bleaching powder): This is used for disinfection of drinking water, stools, urine, sputum. As it decomposes quickly when exposed to air. Solutions must be prepared fresh for each use.
4. Formalin: - A 40% solution is used to disinfect faces, urine and sputum. It is not used for the skin and tissue, as it is an irritant.
5. Tincture of Iodine: - 1-2% iodine is used for cleaning skin and treating injuries to the skin.

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